

Fundamentals of Information Systems Security

Chapter 6 Quiz: Cryptography

Name: _____

Date: _____

Total Points: 100

Time Allowed: 30 Minutes

Instructions

- This quiz contains **20 questions** covering Chapter 6
- Answer all questions to the best of your ability
- Multiple choice questions have only one correct answer
- True/False questions should be marked T or F
- Short answer questions require concise, accurate responses

Section A: Multiple Choice (40 points)

1. (4 points) What is the process of converting plaintext into ciphertext called?
 - A. Decryption
 - B. Encryption
 - C. Hashing
 - D. Encoding
2. (4 points) Which type of cryptography uses the same key for both encryption and decryption?
 - A. Asymmetric cryptography
 - B. Symmetric cryptography
 - C. Hash functions
 - D. Digital signatures
3. (4 points) What is the key distribution problem in symmetric cryptography?
 - A. Keys are too long
 - B. Keys are too short
 - C. Securely sharing the secret key with the intended recipient
 - D. Keys cannot be generated
4. (4 points) Which of the following is an example of a hash function?
 - A. AES
 - B. RSA
 - C. SHA-256

- D. ECC
- 5. (4 points) What is the primary purpose of a hash function?
 - A. Encryption
 - B. Decryption
 - C. Data integrity verification
 - D. Key exchange
- 6. (4 points) Which cryptographic mechanism provides nonrepudiation?
 - A. Symmetric encryption
 - B. Hash functions
 - C. Digital signatures
 - D. Steganography
- 7. (4 points) What does PKI stand for?
 - A. Public Key Infrastructure
 - B. Private Key Infrastructure
 - C. Protected Key Interface
 - D. Public Knowledge Interface
- 8. (4 points) Who issues and verifies digital certificates in PKI?
 - A. Registration Authority (RA)
 - B. Certificate Authority (CA)
 - C. Key Distribution Center (KDC)
 - D. Digital Signing Authority
- 9. (4 points) Which of the following is a symmetric encryption algorithm?
 - A. RSA
 - B. ECC
 - C. AES
 - D. DSA
- 10. (4 points) What is the purpose of a digital signature?
 - A. To encrypt data
 - B. To provide authentication, integrity, and nonrepudiation
 - C. To compress data
 - D. To exchange keys

Section B: True/False (20 points)

- 11. (2 points) Steganography hides the existence of a message.
 - A. True
 - B. False
- 12. (2 points) RSA is a symmetric encryption algorithm.

- A. True
 - B. False
13. (2 points) A one-time pad is theoretically unbreakable.
- A. True
 - B. False
14. (2 points) MD5 is currently considered a secure hash function.
- A. True
 - B. False
15. (2 points) Public keys are kept secret while private keys are shared.
- A. True
 - B. False
16. (2 points) TLS uses asymmetric cryptography to establish a secure session and symmetric cryptography to protect data.
- A. True
 - B. False
17. (2 points) The Certificate Authority (CA) is a trusted entity in PKI.
- A. True
 - B. False
18. (2 points) Quantum cryptography is based on the principles of quantum mechanics.
- A. True
 - B. False
19. (2 points) Post-quantum cryptography is designed to be secure against quantum computers.
- A. True
 - B. False
20. (2 points) HMAC uses a secret key with a hash function.
- A. True
 - B. False

Section C: Short Answer (40 points)

21. (10 points) Explain the difference between symmetric and asymmetric encryption. What are the strengths and weaknesses of each?

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22. (10 points) Describe the four critical goals that cryptography enables in information security.
23. (10 points) What is a digital signature and how does it work? What does it provide?
24. (10 points) Explain the components of Public Key Infrastructure (PKI). What is the role of each component?

End of Quiz